

|                               |                       |                          |
|-------------------------------|-----------------------|--------------------------|
| The judges scoring, note      | Team number:<br>20017 | The judges scoring, note |
| The judges scoring, note<br>、 | Problem:<br>B         | The judges scoring, note |

Title: Books Recommendation

### Abstract

With the development of information technology and the Internet, we are entering gradually from a lack of information era into the era of information overload. Both information consumers and information producers have encountered a lot of challenges, for which recommendation is an important tool to resolve this contradiction, so it is widely used in the products and applications in the Internet .

Question One: The data in the annex user\_book\_score.txt reflects that the book ID and user ID affect book scores. The data in Annex book\_tag.txt reflects that book labels indicates types of books and according to the book ID, we can find the corresponding label, which affects user ratings of books . In Annex user\_read\_history.txt , the history users viewed books reflects the kind of books the users like, thus affecting the score. In Annex user\_social.txt, concern users to their friends reflects the relationship between the user 's choice and liking of books. Thus contact between the user and the books are drawn, which affects the score. So the five factors impacting the user rating for the books are: book ID, book labels, book history users browsing, users concerned friends and the user ID.

Question Two: the user score of books in the annex predict.txt considers the factors in user score of books. First of all, data are in kmo test by application of software spss, and the test sig <0.05, so principal component analysis of the data is done. In order to ensure the reliability of the results, data for books ID and user ID are standardized, and five main indicators are in component analysis. When three main components are acquired, the cumulative contribution rate reaches 81.371%. These three main components are used to do factor analysis, and integrated model for principal component is:  

$$F = 0.397F_1 + 0.33F_2 + 0.273F_3$$
. Prediction for user ratings for books in predict.txt is ultimately got.

Question Three: through the book history the user browsing and analysis of the book scores, on the one hand the similarity of book labels are found in books browsing history to further infer the user's favorite kind of books, and on the other hand readers' favorite books can also be judged by the similarity high score books on the rating scale. Considering the two aspects of information above, collaborative filtering recommendation algorithm is used to calculate the similarity between user ID to elect two or three books which have the most comprehensive similarity and recommend to the users.

Keywords: overlap、 similarity、 principal component analysis、 、 factor analysis、 collaborative filtering algorithm

## 1 PROBLEM RESTATED

With the development of information technology and Internet, people gradually went out into an information overloaded world. At this point, whether information consumers or information producers are experiencing great challenges: As for information consumers, it is a very difficult task to find the interesting information from a large amounts of information. For information producers, letting their own production of information stand out and drawing user's attention would also be a very tough issue.

Recommendation is a key tool to resolve this contradiction. The products and applications are widely used on the Internet, including the relevant search, recommended topic, various products in electronic commerce recommendation, dating and recommendation on the social network.

We obtained a well-known user behavior information online bookstores, including rating statistics for books, tag information and the user's social relationships. Please complete the following questions based on the data.

- 1 Analyzing the factors affecting the assessment of books;
- 2 Develop a model, predicting book scores in the attachment of predict.txt;
- 3 Aiming predict.txt Annex users, we recommend three books to users who did not read these books before.

## 2 PROBLEM ANALYSIS

Question One:

To analyze the impact of the book users rating factors, it introduces the annex book\_tag.txt, user\_book\_score.txt, user\_read\_history.txt, user\_social.txt data to the software, after data processing, data attachments can be seen in user\_book\_score.txt book ID and the user ID of the main factors affecting the score. After Annex book\_tag.txt, user\_social.txt user\_read\_history.txt the data analysis, the label represents different types of books, thus affecting the score. Users can find the labels of various kinds of books, looking through the history books of the ID according to books with different ID. For users concerning about the number of friends ID, you can analyze the relationship between the obtained degree of users' choices and favorite books, and finally draw contact between the users and the books and affect the score.

Question Two:

To require users to predict the score of books in predict.txt annex, it must be based on a known problem affecting users of books score factors and use principal component analysis modeled by principal component analysis:

Considering the factors in the user assessment of books. First of all, the data using software kmo test, the test sig <0.05, principal component analysis of the data. As the book ID, user ID is just a symbol in order to ensure reliability of the results,

then the need to standardize the raw index data processing , composition analysis of five indicators shots , when acquired three main components , the cumulative contribution rate of 81.371% , and using these three principal components factor analysis , principal component solution was eventually integrated model is:

$$F = 0.397F_1 + 0.33F_2 + 0.273F_3 \text{ ultimately predict predict.txt user ratings for books .}$$

Question three:

After the user viewing the history books as well as analysis of the scores of books , on the one hand, we find the similarity of books tagged books from browsing history and infer users' favorite books .On another hand, finding the rating scale high level similarity of books can be judged. Considering the above two aspects of information ,with the usage of collaborative filtering recommendation algorithm, we calculate the similarity between the users' ID to elect the most comprehensive similarity among three books recommended to the user .

### 3. MODEL ASSUMPTIONS

- 1 Assumes that the data follow a normal distribution .
- 2 Various factors are independent each other .
- 3 Assuming characteristic values can be seen in a way is a principal component of impact indicators efforts size.
- 4 . Ignore the overall user ratings for books errors based on limited sample data.

### 4 SYMBOL DESCRIPYTION

|       |                            |
|-------|----------------------------|
| $y$   | User data rates            |
| $x_1$ | Books ID standardization   |
| $x_2$ | Number of Friends          |
| $x_3$ | User ID standardization    |
| $x_4$ | The number of books labels |
| $x_5$ | Users browsing history     |
| $F_1$ | Principal component 1      |
| $F_2$ | Principal component 2      |
| $F_3$ | Principal component 3      |

|                  |                                                        |
|------------------|--------------------------------------------------------|
| $F$              | Principal component                                    |
| $I_{i,j}$        | Common users and excessive collection of items comment |
| $R_{i,c}$        | User ratings for books                                 |
| $\overline{R_i}$ | Average user rating of books                           |
| $\overline{R_j}$ | Average user rating of books                           |

## (5) ESTABLISH AND SOVEL THE MODEL

### 5.1 Question One:

#### 5.1.1 Establishment of a model :

According to the data given in the title meaning and accessories , we analyzes Annex user\_book\_score.txt user ID, books ID, listing three columns of data , the first stage is divided into three contact:

Contacts

- ( 1 ) the connection between books and books ID
- ( 2 ) the connection between the user ID of the user ID ;
- ( 3 ) the connection between the user ID and the ID of books .

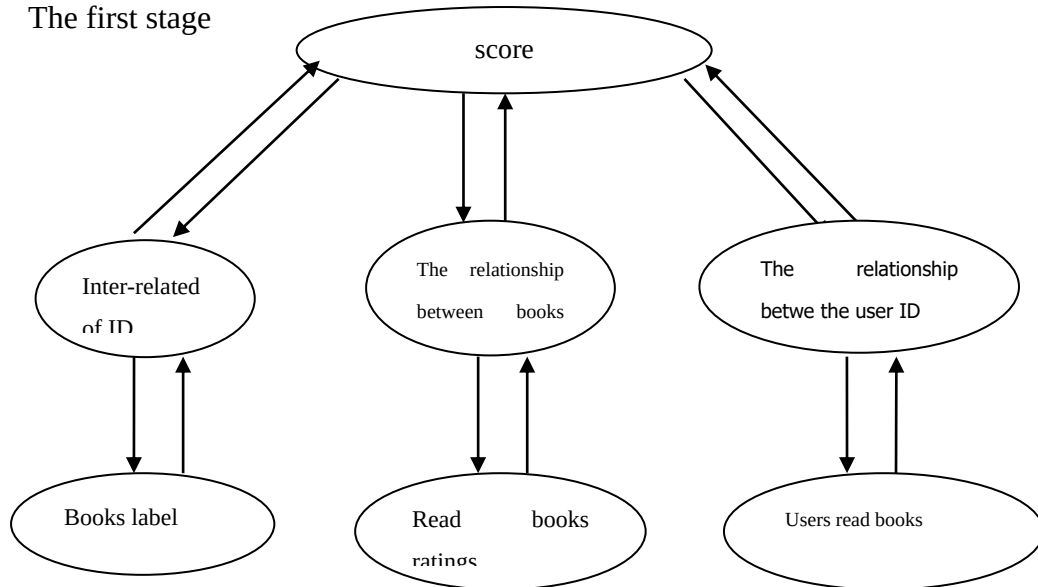
The second level given in Annex book\_tag.txt books label data further analysis:

( 1 ) For users who have same hobbies , selected books will be similar, so the degree of overlap would be high, books overlap size comparison between the labels can determine the similarity between the book and the book size ;

( 2 ) Then according to Annex user\_read\_history.txt user viewed history books ID  
Data Analysis: Comparison of the similarity between the user read the book can determine the size of contact between the users.

( 3 ) Analyzing data of Annex user\_book\_score.txt, according to user browsing through the books of history and scoring the level of similarity, it can be brought back to the size of the similarity between the books and label analysis , we can let the users to select the degree and interests of books due to the relationship between the obtained contact between the users and books . Conclusion can be drawn on the second stage by the first stage, which means analysis of the impact of the user i.e. book rating factors.

The first stage



The second stage

Figure 5-1 score affect users on books analysis chart

Figure 5-1 Effect create user rating analysis diagram Books , influence the relationship between the book ID label overlap books , books overlap book tag ID to the relationship between the reaction , the relationship between the book ID and the interaction score ; the relationship between books and user ID affect the user 's score read books , read books user ratings counterproductive to the relationship between books and user ID , user ID between books and the relationship between score interaction ; affect the relationship between the user ID users read books , read books on the user reaction to the relationship between a user ID .

5.1.2 Solving a model :

Figure 4-1 affected by user rating analysis chart books analyze the impact on users of books score factors :

- ( 1 ) ID books
- Tag number ( 2 ) books
- ( 3 ) User ID history books viewed
- ( 4 ) the number of users concerned friends ID
- ( 5 ) the user ID

5.1.2 Solving a model :

Figure 4-1 affected by user rating analysis chart books analyze the impact on users of books score factors :

- ( 1 ) ID books
- Tag number ( 2 ) books
- ( 3 ) User ID history books viewed
- ( 4 ) the number of users concerned friends ID
- ( 5 ) the user ID

## 5.2 Second problem:

### 5.2.1 Establishment of two models :

Principal component analysis : dimensionality reduction aims to use the idea of the multi-index into a few comprehensive index . It is given to a group of related variables are not converted into another group of related variables by linear transformation , the new variables are sequentially arranged in order of decreasing variance . Hold variable in the same mathematical transformation of the total variance , the first variable having the largest variance is called the first principal component , a large variance of the second sub- variables, and is not related to the first variable , referred to as the second component , and so on .

. The aim is to use less data variables to explain most of the original variables, many of the high correlation of variables into our hands independent of each other or related variable .

Factor analysis: from the study of internal variables related dependencies departure , some variables have complicated relations attributed to a few factors integrated multivariate statistical analysis methods. The basic idea is to classify the observed variables , the correlation is high, that is relatively close contact points in the same class , but not the same correlation between variables is low, then the variable is actually representative of each category a basic structure, that common factor .

First of all data kmo testing, achieved through software spss obtain Table 5-1kmo examination table .

Chart 5-1 KMO Inspection Table

|                                                  |      |       |
|--------------------------------------------------|------|-------|
| Kaiser-Meyer-Olkin Measure of Sampling Adequacy. |      | .235  |
| Approx. Chi-Square                               |      | 7.349 |
| Bartlett's Test of Sphericity                    | df   | 10    |
|                                                  | Sig. | .000  |

By kmo test , sig <0.05, the data can be the principal component analysis .

By principal component analysis , principal component analysis construct mathematical models :

$$F_m = \sum_{m=1}^5 a_m x_m \quad m \in \{m \in N, 1 \leq m \leq 5\}$$

Which  $a_m$  is characterized by the covariance matrix of  $x_m$  values corresponding

eigenvectors are normalized after the original variable values  $x_m$  .

### 1 Data Standardization:

In the evaluation of a number of indicators , due to book ID, user ID , and only a symbol ID friends , with different dimensions and when the magnitude of the

difference between the level of each index is large , if the index values directly using the original data for analysis, will highlight the role of high value targets in the comprehensive analysis , the relative weakening of the role of values lower level indicators. Therefore, in order to ensure reliability of the results , the need for the raw data are normalized indicators .

By the formula  $\frac{X - \min}{\max - \min}$  Of books ID, user ID two types of data standardization

( 1 ) book ID :min=100009,max= 999707.

( 2 ) User ID :min= 100035,max= 9999804.

2. determine the impact of the correlation coefficient matrix and calculate the correlation between the factors :

The import data into the software spss, the book ID, user ID standardization factor analysis to determine the correlation coefficient obtained in Table 5-2 Matrix .

Chart 5-2 Determine the correlation coefficient matrix

|                 |                                             | Standar<br>dized<br>books<br>ID | The number<br>of<br>concerned<br>friends ID | The<br>number<br>of books<br>labels | User ID<br>standard<br>ization | Users<br>browsing<br>history |
|-----------------|---------------------------------------------|---------------------------------|---------------------------------------------|-------------------------------------|--------------------------------|------------------------------|
| Correlatio<br>n | Standardize<br>d books ID                   | 1.000                           | .144                                        | -.183                               | -.258                          | -.005                        |
|                 | The number<br>of<br>concerned<br>friends ID | .144                            | 1.000                                       | .089                                | .548                           | .036                         |
|                 | The number<br>of books<br>labels            | -.183                           | .089                                        | 1.000                               | -.315                          | .304                         |
|                 | User ID<br>standardizat<br>ion              | -.258                           | .548                                        | -.315                               | 1.000                          | -.035                        |
|                 | Users<br>browsing<br>history                | -.005                           | .036                                        | .304                                | -.035                          | 1.000                        |

3 Select the number of principal components principles:

( 1 ) the cumulative contribution rate of 70 % to 80% ;

( 2 ) the formal m principal components  $z_1, z_2, \dots, z_m$  of the original variable  $x_i$

contribution rate is not too small .

Data analysis can be obtained from the characteristic value in Table 4-3 , the variance , the variance accumulation , load table , the impact of various rating index corresponding feature value

$\lambda_1 = 1.617$ ,  $\lambda_2 = 1.342$ ,  $\lambda_3 = 1.110$ ,  $\lambda_4 = 0.754$ ,  $\lambda_5 = 0.178$ . Number of principal components extracted main component corresponds to the principle of eigenvalues greater than 1 the former principal components m ; cumulative contribution rate of 81.371% , ie standardization books ID, user ID concerned about the number of friends , the label as the number of books main ingredients. And to each of the main components of the corresponding eigenvalues extracted principal components accounted for the total sum of the eigenvalues as weights to calculate the proportion of principal components integrated model.

4 . Principal component expression:

Chart 5-3 Characteristic value , variance, variance cumulative load table

| Component | Initial Eigenvalues |               |              | Extraction Sums of Squared Loadings |               |              | Rotation Sums of Squared Loadings |               |              |
|-----------|---------------------|---------------|--------------|-------------------------------------|---------------|--------------|-----------------------------------|---------------|--------------|
|           | total               | % of Variance | Cumulative % | total                               | % of Variance | Cumulative % | total                             | % of Variance | Cumulative % |
| 1         | .617                | 32.30         | 32.30        | .617                                | 32.30         | 32.30        | .558                              | 31.152        | 31.152       |
| 2         | .342                | 18.46         | 50.76        | .342                                | 18.46         | 50.76        | .375                              | 20.498        | 51.250       |
| 3         | .110                | 5.83          | 56.59        | .110                                | 5.83          | 56.59        | .136                              | 7.421         | 58.671       |
| 4         | .075                | 4.07          | 60.66        |                                     |               |              |                                   |               |              |
| 5         | .018                | 0.95          | 61.61        |                                     |               |              |                                   |               |              |

$$F = \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} F_1 + \frac{\lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} F_2 + \frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} F_3 \quad (3)$$

5.2.2 The Solving to The Second Model :

Appendix Table 1 will import SPSS software item factor analysis , principal component coefficient matrix can be obtained as shown in Table 5-4 principal component coefficient matrix.

Chart 5-4 Principal Component Matrix

**Component Score Coefficient Matrix**

|                                | Component |       |        |
|--------------------------------|-----------|-------|--------|
|                                | 1         | 2     | 3      |
| User ID standardization /x1    | -0.613    | -.322 | 0.864  |
| Users browsing history/x2      | 0.562     | .352  | 0.408  |
| Books ID standardization /x3   | -0.359    | .634  | .032   |
| Number of Friends /x5          | 0.719     | .164  | -0.167 |
| The number of books labels /x5 | -0.185    | .587  | 0.241  |

Get the main component expression:

$$F_1 = -0.613x_1 + 0.562x_2 - 0.359x_3 + 0.719x_4 - 0.185x_5 \quad (4)$$

$$F_2 = -0.322x_1 + 0.352x_2 + 0.634x_3 + 0.164x_4 + 0.587x_5 \quad (5)$$

$$F_3 = 0.864x_1 + 0.408x_2 + 0.032x_3 - 0.167x_4 + 0.241x_5 \quad (6)$$

Corresponding to each of the main components of the characteristic features of the total value of the principal component and the value representing the ratio of the extracted principal components as weights to calculate integrated model.

$$F = \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} F_1 + \frac{\lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} F_2 + \frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} F_3 \quad (7)$$

Table 5-4 winner components corresponding to the characteristic value into ( 7):

$$F = 0.397F_1 + 0.33F_2 + 0.273F_3 \quad (8)$$

By schedule 1 ,( 4 ) , ( 5 ) , ( 6 ) and ( 8 ) can be obtained score values, Chart 5-5 users that score table books.

Chart 5-5 User rating table for books

| User ID | Books ID | Score | Accurate Ratings |
|---------|----------|-------|------------------|
|---------|----------|-------|------------------|

|         |        |       |   |
|---------|--------|-------|---|
| 7245481 | 794171 | 4. 04 | 4 |
| 7245481 | 381060 | 5. 00 | 5 |
| 7245481 | 776002 | 4. 92 | 5 |
| 7245481 | 980705 | 4. 62 | 5 |
| 7245481 | 354292 | 4. 55 | 5 |
| 7245481 | 738735 | 4. 65 | 5 |
| 7625225 | 473690 | 3. 26 | 3 |
| 7625225 | 929118 | 3. 02 | 3 |
| 7625225 | 235338 | 3. 19 | 3 |
| 7625225 | 424691 | 3. 26 | 3 |
| 7625225 | 916469 | 3. 22 | 3 |
| 7625225 | 793936 | 3. 15 | 3 |
| 4156658 | 175031 | 2. 52 | 3 |
| 4156658 | 422711 | 2. 84 | 3 |
| 4156658 | 585783 | 2. 24 | 2 |
| 4156658 | 412990 | 2. 25 | 2 |
| 4156658 | 134003 | 2. 54 | 3 |
| 4156658 | 443948 | 2. 58 | 3 |
| 5997834 | 346935 | 4. 47 | 4 |
| 5997834 | 144718 | 3. 60 | 4 |
| 5997834 | 827305 | 3. 14 | 3 |
| 5997834 | 219560 | 3. 89 | 4 |
| 5997834 | 242057 | 3. 51 | 4 |
| 5997834 | 803508 | 2. 61 | 3 |
| 9214078 | 310411 | 4. 58 | 5 |
| 9214078 | 727635 | 4. 16 | 4 |
| 9214078 | 724917 | 3. 63 | 4 |
| 9214078 | 325721 | 4. 43 | 4 |
| 9214078 | 105962 | 4. 16 | 4 |
| 9214078 | 235338 | 4. 44 | 4 |
| 2515537 | 900197 | 3. 65 | 4 |
| 2515537 | 680158 | 3. 00 | 3 |
| 2515537 | 770309 | 3. 37 | 3 |
| 2515537 | 424691 | 3. 56 | 4 |
| 2515537 | 573732 | 2. 79 | 3 |
| 2515537 | 210973 | 3. 36 | 3 |

### 5.3 Question Three:

#### 5.3.1The Establishment of the Third model:

The establishment of the third model:

Collaborative Filtering Recommendation Algorithm: By analyzing users' interests, it find the specified users who have similar interests in the user group.. Sum the evaluation of information of these similar users then come into being the specified

user preferences to the information system prediction. The advantage of this algorithm: First it can filter the information that we can't use machine to analyze. Second, it can also filter some information that based on some complicated and inexpressible conceptions. Third, the novelty of the recommendation.

Channeling that user\_book\_score.txt, book\_tag.txt, user\_read\_history.txt into the software Excel and getting the score chart, label chart and the browsing history chart. First, you should find the book ID that has the highest score in the books that haven't been browsed by people and that have been browsed by people. Second, you should find those labels that have the highest score in the label chart. Through comparing the degree of overlap of the label between them, recommend the user three books that have the biggest similarity with the books that have been browsed in the books that haven't been seen.

By anglicizing the user viewed the history books as well as the scores of books, on the one hand, we can find the similarity of books tagged books from browsing history , you can infer what the user favorite books ; hand found on the rating scale score high level similarity of books can be judged reader favorite books . considering the above two aspects of information , the use of collaborative filtering recommendation algorithm to calculate the similarity between the user ID to elect the most comprehensive similarity between two three books recommended to the user .

Collaborative filtering recommendation of three steps between the user ID:

( A ) calculating the similarity between the user ID :

We can be related to the similarity , let users  $i$  and users  $j$  grade common assessment . Using the *Pearson* correlation coefficient measure , assuming the user  $i$  and the users  $j$  assessment by the excessive collection of items  $I_{i,j} = I_i \cap I_j$  .

The user  $i$  and the user  $j$  similarity  $Sim(i, j)$  measure by the *Pearson* correlation coefficient:

$$Sim(i, j) = \frac{\sum_{c \in I_{i,j}} (R_{i,c} - \bar{R}_i)(R_{j,c} - \bar{R}_j)}{\sqrt{\sum_{c \in I_{i,j}} (R_{i,c} - \bar{R}_i)^2} \sqrt{\sum_{c \in I_{i,j}} (R_{j,c} - \bar{R}_j)^2}}$$

$R_{i,c}$  indicates that the user  $i$ 's score of books ,  $\bar{R}_i$  and  $\bar{R}_j$  represent the user  $i$  and the user  $j$ 's average score of books .

( 2 ) Nearest Neighbor Discovery

Analysis of the user to find a collection of his descending order of similarity , is set to N. N data collection arranged by the similarity of the size of the elements most closely identify , analyze a set of elements closest to select the first N similarity largest user user as a neighbor .

### ( 3 ) Recommended Set Produced

After the target user's " nearest neighbor " sets to produce , we can calculate two types of results : the predicted value of users' interests in any item and the recommendation set in the form of Top – N. The predicted value of the target user  $i$  's interest in the books item  $j$  is calculated by:

$$P_{i,j} = \frac{\sum_{m=1}^n sim(i, m) \times (R_{m,j} - \overline{R_m})}{\sum_{m=1}^n sim(i, m)}$$

Which  $sim(i, m)$  represents the similarity between the user  $i$  and the nearest

neighbor set of users  $m$  .  $R_{m,i}$  is a user  $i$  score of books  $j$  ,  $\overline{R_i}$  and  $\overline{R_m}$  represent

users and the average score of books ,  $n$  is the number of nearest neighbors .

#### 5.3.2 The Solving to the Third Model :

By analysing of the users' viewed books' history as well as the scores of books : one to find the similarity of books tagged books from browsing history , you can infer what readers love books ; hand found on the rating scale score high level similarity of books can be judged reader favorite books . according to a three-step formula collaborative filtering recommendation , combined with the processing of data to find the bookmark similarity , it can be recommended to the user based on the similarity to identify the three books .

Through the software, Excel, ratings , labels . , Browsing history data processing , first for the user ID is 7245481 , and seek , the user does not read books as 929118,438265,172132 ID tag with the user ID tag read books regarded as highly relevant , namely books recommended to the user ID for 929118,438265,172132 these three books.

Similarly available to the user ID for the user 7625225,4156658,5997834,9214078,2515537 Recommended Books Recommended

Chart 5-6 Recommended Books

| books<br>ID<br>numbe | 7245481 | 7625225 | 4156658 | 5997834 | 9214078 | 2515537 |
|----------------------|---------|---------|---------|---------|---------|---------|
| 1                    | 929118  | 900197  | 976224  | 346935  | 224635  | 300020  |

|   |        |        |        |        |        |        |
|---|--------|--------|--------|--------|--------|--------|
| 2 | 438265 | 827305 | 437064 | 770814 | 849294 | 194845 |
| 3 | 172132 | 275636 | 456400 | 184158 | 890399 | 634885 |

## 6 . Promotion model

Collaborative filtering recommendation of this model can also be applied to the film site. The site platform consists of two parts: sever and client. By monitoring the film browsing history and score of user comment through user ID, the sever analyses the interests of users, and then recommend suitable films for users. The client provide the interaction between client and system in the form of browser, make labels and comments for the films through the favorites.

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Appendix:

The Chart Of Data Normalization

| User ID | The number of concerned friends ID | Users concerned about the friend ID number | Standard ization of books ID | Books ID | the number of standar dized Bookm ark | The nu mbe r of boo ks labe ls | The number of books |
|---------|------------------------------------|--------------------------------------------|------------------------------|----------|---------------------------------------|--------------------------------|---------------------|
| 7245481 | 0.074626866                        | 6                                          | 0.771550009                  | 794171   | 0.830250719                           | 19                             | 0.771580883         |
| 7245481 | 0.074626866                        | 6                                          | 0.312383711                  | 381060   | 2.885326757                           | 34                             | 0.312396211         |
| 72454   | 0.074626                           | 6                                          | 0.75135                      | 77600    | 2.6113                                | 32                             | 0.75138552          |

|             |                 |    |                 |            |                  |    |                 |
|-------------|-----------------|----|-----------------|------------|------------------|----|-----------------|
| 81          | 866             |    | 5455            | 2          | 16619            |    |                 |
| 72454<br>81 | 0.074626<br>866 | 6  | 0.97887<br>958  | 98070<br>5 | 1.9262<br>91273  | 27 | 0.97891874<br>9 |
| 72454<br>81 | 0.074626<br>866 | 6  | 0.28263<br>1505 | 35429<br>2 | 1.6522<br>81134  | 25 | 0.28264281<br>5 |
| 72454<br>81 | 0.074626<br>866 | 6  | 0.70993<br>3778 | 73873<br>5 | 1.9262<br>91273  | 27 | 0.70996218<br>6 |
| 76252<br>25 | 0.343283<br>582 | 24 | 0.41534<br>0481 | 47369<br>0 | 2.3373<br>0648   | 30 | 0.41535710<br>1 |
| 76252<br>25 | 0.343283<br>582 | 24 | 0.92154<br>1451 | 92911<br>8 | 0.6932<br>4565   | 18 | 0.92157832<br>6 |
| 76252<br>25 | 0.343283<br>582 | 24 | 0.15041<br>6028 | 23533<br>8 | 1.7892<br>86204  | 26 | 0.15042204<br>7 |
| 76252<br>25 | 0.343283<br>582 | 24 | 0.36087<br>8873 | 42469<br>1 | 2.3373<br>0648   | 30 | 0.36089331<br>3 |
| 76252<br>25 | 0.343283<br>582 | 24 | 0.90748<br>2289 | 91646<br>9 | 2.2003<br>01411  | 29 | 0.90751860<br>1 |
| 76252<br>25 | 0.343283<br>582 | 24 | 0.77128<br>881  | 79393<br>6 | 1.6522<br>81134  | 25 | 0.77131967<br>3 |
| 41566<br>58 | 0.283582<br>09  | 20 | 0.08338<br>5758 | 17503<br>1 | 0.1452<br>25373  | 14 | 0.08338909<br>5 |
| 41566<br>58 | 0.283582<br>09  | 20 | 0.35867<br>8134 | 42271<br>1 | 2.6113<br>16619  | 32 | 0.35869248<br>7 |
| 41566<br>58 | 0.283582<br>09  | 20 | 0.53993<br>0065 | 58578<br>3 | -1.7728<br>45595 |    | 0.53995167<br>1 |
| 41566<br>58 | 0.283582<br>09  | 20 | 0.34787<br>3398 | 41299<br>0 | -1.7728<br>45595 |    | 0.34788731<br>8 |
| 41566<br>58 | 0.283582<br>09  | 20 | 0.03778<br>379  | 13400<br>3 | 0.2822<br>30443  | 15 | 0.03778530<br>2 |
| 41566<br>58 | 0.283582<br>09  | 20 | 0.38228<br>2722 | 44394<br>8 | 0.6932<br>4565   | 18 | 0.38229801<br>9 |
| 59978<br>34 | 0.029850<br>746 | 3  | 0.27445<br>4317 | 34693<br>5 | 3.9813<br>67311  | 42 | 0.27446529<br>9 |
| 59978<br>34 | 0.029850<br>746 | 3  | 0.04969<br>3342 | 14471<br>8 | 2.3373<br>0648   | 30 | 0.04969533      |
| 59978<br>34 | 0.029850<br>746 | 3  | 0.80837<br>7922 | 82730<br>5 | 1.6522<br>81134  | 25 | 0.80841027      |
| 59978<br>34 | 0.029850<br>746 | 3  | 0.13287<br>9033 | 21956<br>0 | 2.8853<br>26757  | 34 | 0.13288435      |
| 59978<br>34 | 0.029850<br>746 | 3  | 0.15788<br>409  | 24205<br>7 | 2.2003<br>01411  | 29 | 0.15789040<br>8 |
| 59978<br>34 | 0.029850<br>746 | 3  | 0.78192<br>7936 | 80350<br>8 | 0.6932<br>4565   | 18 | 0.78195922<br>5 |
| 92140       | 0.134328        | 10 | 0.23385         | 31041      | 2.0632           | 28 | 0.23386783      |

|             |                 |    |                 |            |                 |    |                 |
|-------------|-----------------|----|-----------------|------------|-----------------|----|-----------------|
| 78          | 358             |    | 8473            | 1          | 96342           |    |                 |
| 92140<br>78 | 0.134328<br>358 | 10 | 0.69759<br>6305 | 72763<br>5 | 1.3782<br>70996 | 23 | 0.69762421<br>9 |
| 92140<br>78 | 0.134328<br>358 | 10 | 0.69457<br>5291 | 72491<br>7 | 0.4192<br>35512 | 16 | 0.69460308<br>4 |
| 92140<br>78 | 0.134328<br>358 | 10 | 0.25087<br>5294 | 32572<br>1 | 1.7892<br>86204 | 26 | 0.25088533<br>2 |
| 92140<br>78 | 0.134328<br>358 | 10 | 0.00661<br>6665 | 10596<br>2 | 1.2412<br>65927 | 22 | 0.00661692<br>9 |
| 92140<br>78 | 0.134328<br>358 | 10 | 0.15041<br>6028 | 23533<br>8 | 1.7892<br>86204 | 26 | 0.15042204<br>7 |
| 25155<br>37 | 0.029850<br>746 | 3  | 0.88939<br>622  | 90019<br>7 | 2.6113<br>16619 | 32 | 0.88943180<br>9 |
| 25155<br>37 | 0.029850<br>746 | 3  | 0.64482<br>6375 | 68015<br>8 | 1.3782<br>70996 | 23 | 0.64485217<br>8 |
| 25155<br>37 | 0.029850<br>746 | 3  | 0.74502<br>7776 | 77030<br>9 | 2.0632<br>96342 | 28 | 0.74505758<br>8 |
| 25155<br>37 | 0.029850<br>746 | 3  | 0.36087<br>8873 | 42469<br>1 | 2.3373<br>0648  | 30 | 0.36089331<br>3 |
| 25155<br>37 | 0.029850<br>746 | 3  | 0.52653<br>5571 | 57373<br>2 | 0.9672<br>55788 | 20 | 0.52655664      |
| 25155<br>37 | 0.029850<br>746 | 3  | 0.12333<br>4719 | 21097<br>3 | 1.9262<br>91273 | 27 | 0.12333965<br>4 |